Rui Fazendeiro

Electrical Systems & Software Engineering | CAD & Systems Developer

📍 Baião, Portugal 📩 ruifazendeiro@outlook.pt 💼 linkedin.com/in/rfazendeiro 💻 github.com/RuiFazendeiro

PROFILE

Highly skilled **CAD/ECAD Designer** and **Computer Engineering student** specializing in the convergence of electrical systems, industrial automation, and software development. Extensive experience in technical drawings for electrical substations and MV/LV switchboards, combined with a proven track record of developing custom software solutions, embedded systems (IoT), and process automation tools to optimize industrial engineering workflows.

EDUCATION & CREDENTIALS

BSc Student in Computer Engineering | Universidade Aberta (UAb) *2024 – Present (Expected 2028)*

Licensed Electrotechnical Technician (Level 4 QNQ/QEQ) | ANQEP / DGEG Registration

Passed with Distinction

- Official qualification certified for electrical installations, automation systems, and electrical machine testing. Enables formal technical responsibility up to 10.35 kVA.

TECHNICAL SKILLS

Automation & Embedded Systems: C++, Firmware Development (ESP32/STM32), Modbus RS485, Edge Computing, IoT Architecture, FreeRTOS.

Software & Databases: Python (Streamlit), SQL, Relational Databases (Access), Git version control.

Industrial CAD/ECAD & Geospatial: AutoCAD, SolidWorks (Parametric Design & Jigs), Revit (BIM), CYPE, ArcGIS.

Testing & Quality Assurance: Instrumentation (Agilent loggers, Keysight/Fluke multimeters), Accelerated Environmental Testing, Root-Cause Analysis, Lean Manufacturing (5S).

PROFESSIONAL EXPERIENCE

Electrotechnical CAD Designer | Matelfe *2025 – Present*

- Develop detailed CAD technical layouts and multi-line electrical schematics for medium/low voltage electrical transformer stations (substations) and industrial switchboards according to utility specifications.
- **Workflow Automation:** Concepted and deployed a custom local application using Python, Streamlit, and SQL to index and query metadata from hundreds of complex CAD drawing variations, cutting engineering configuration bottlenecks.

CAD Designer & Topography Technician | BC Engenharia *2024 – 2025*

- Conducted field topography surveys, engineering data acquisition, and spatial asset mapping using ArcGIS.
- Produced high-precision structural documentation, technical cross-sections, elevations, and layout specifications using AutoCAD, Revit, and CYPE.
- Collaborated within cross-functional teams to validate architectural limits and civil infrastructure metrics.

Engineering Assistant | MyProject by Casa Gomes Engenharia *2022 – 2023*

- Assisted in official building energy certification processes, performing technical thermal behavior analysis and compliance benchmarking under national regulations.
- Developed comprehensive architectural, HVAC, and internal distribution electrical CAD schematics for residential and commercial layouts.

Field POS IT Technician | Verifone *2021 – 2022*

- Managed on-site installation, technical commissioning, and operational deployment of financial transaction hardware and Point of Sale (POS) infrastructures.
- Executed component-level electronic diagnostics, board repairs, and localized hardware/software troubleshooting.
- Performed field firmware updates and network connectivity adjustments under strict client SLA performance thresholds.

Laboratory Electrical Technician & CAD Designer | Yazaki Europe *2016 – 2019*

- Executed electrical validation and environmental stress compliance tests on automotive wiring harness architectures within the Central R&D / D&D department.
- Monitored variables using specialized instrumentation (Agilent data loggers, Keysight/Fluke bench multimeters) under accelerated aging profiles (thermal shock, vibration, humidity, climate chambers).
- **Supplementary Jigs Design:** Engineered and modeled mechanical testing fixtures and calibration validation matrix jigs completely from scratch using SolidWorks, enforcing rigid OEM manufacturer standards.

KEY ENGINEERING PROJECTS

Smart Electrical Panel – IoT & Edge Computing

Professional Level 4 Technical Project (Passed with Distinction)

- Transformed a passive electrical distribution board into an active edge-computing nodes node using an ESP32-S3 microcontroller and C++ firmware. Implemented multi-channel telemetry validation via Modbus RS485, closed-loop control with real-time state machines, dynamic load shedding algorithms to eliminate grid trip faults, and isolated phase-control switching safely on DIN rails.

Parametric Compact Substation Design (SolidWorks)

Industrial Mechanical & Product Engineering Design

- Engineered a fully parametric 3D structural model of a compact distribution transformer station using global variables and geometric equations in SolidWorks. Dynamic rebuilding logic allows for rapid custom dimensioning and automated BOM generation, cutting configuration setup workflows by 70% while verifying clearance limits.